03. Root Finding: Newton-Raphson

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Last time: Probability Integral Transform

if $Z \sim F(t)$ Then $\boxed{U = F(Z)} \sim Unif[0,1]$

$Z = F^{-1}(U) \sim F(t)$

Problem: Sometimes we can't find F^{-1} in a closed formula.

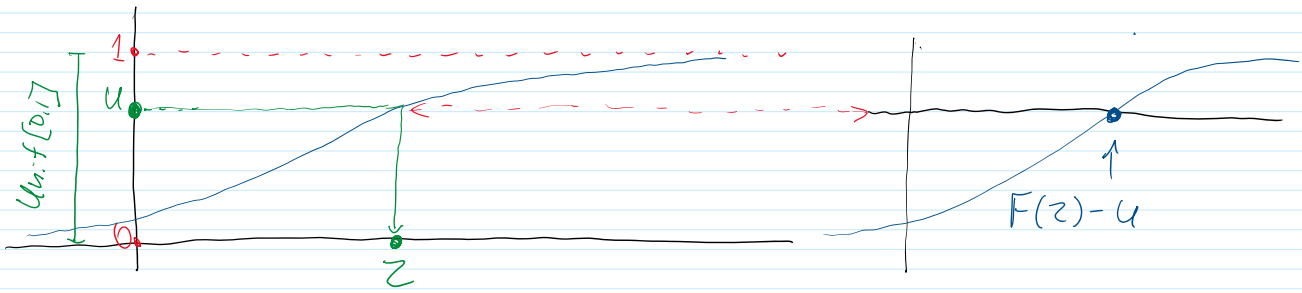
E.G. Normal$(0,1)$ then $F(t) = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx$

- No formula for F^{-1} even though F is continuous, differentiable, strictly increasing, etc

Reimagine "inversion" as "root-finding"

$$\boxed{0 = F(Z) - U}$$

Algorithmically:
- Generate $U \sim Unif[0,1]$
- Find root in Z of $F(Z) - U$

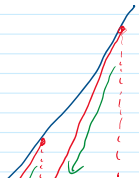


Newton-Raphson

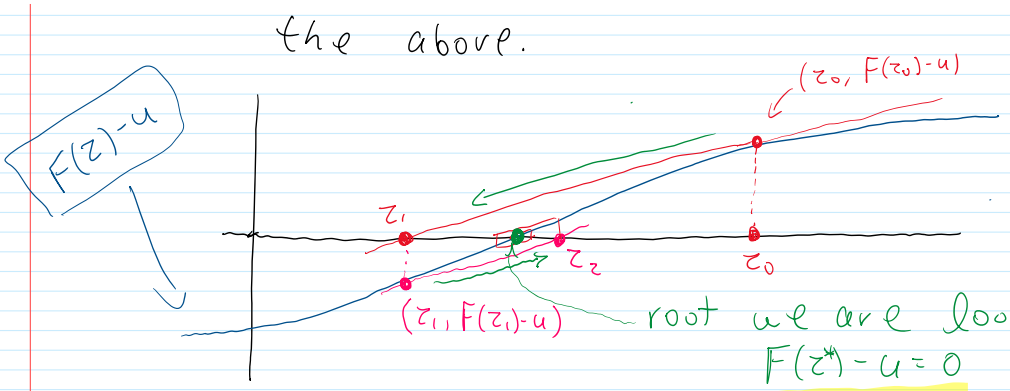
Besides a CDF $F(t)$, we need the derivative $f(t)$ which we assume exists. $\frac{d}{dt}F(t)$

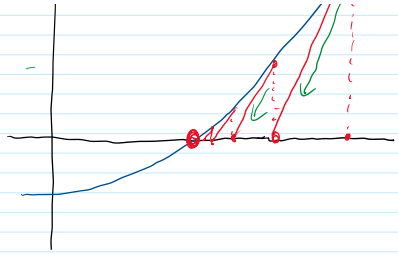
- pick a starting value z_0 in $\mathbb{R}$
- Draw a tangent line at $(z_0, F(z_0) - u)$
- Follow this line to the horizontal axis
  - ⇒ get a new point z_1 and repeat the above.

$(z_0, F(z_0) - u)$



the above.





Tangent line equation is ...

$$f(z_0) = \frac{(F(z_0)-u) - 0}{z_0 - z_1}$$

← change in vertical

← change in Horizontal

slope derivative

unknown

Rearrange the equation: $z_1 = z_0 - \underbrace{\frac{F(z_0)-u}{f(z_0)}}$

new value — old value — original — derivative — update based on the derivative.

Will this always work?

- Warning! we are dividing by the derivative / PDF
  What if this is really big or really small?

When does it work really well?

- z_0 is close to the root, we converge quickly
- Smooth functions are locally linear